Comparative Analysis of Deep Learning Models in Image Captioning in Bangla

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***Abstract*—Image captioning is an important problem in computer vision with wide applications. In this study, we conduct a comparative analysis of popular deep-learning models for automatic image captioning in Bangla language. We experiment with CNN-RNN, Attention-based, Transformer-based and GRU based models and train them on a Bangla image captioning dataset consisting of 8,743 images. The models are evaluated based on BLEU, METEOR, ROUGE-L and CIDEr scores. Our results show that the Transformer model achieves the best performance with average scores of BLEU-4: 46, METEOR: 48, ROUGE-L: 47 and CIDEr: 55, outperforming other models. We believe this work will help advance research on Bangla image captioning.**

I. Introduction

Image captioning is the task of automatically generating linguistic descriptions of image content. It has many applications such as assistive technologies for content search (Pan et al. 2016; Yu et al. 2016). Recent advances in deep learning have led to significant progress in English image captioning (You et al. 2016; Donahue et al. 2015). However, the Bangla captioning of images remains relatively less developed.

Bangla is one of the most widely spoken languages, yet access to multimedia content in Bangla can be limited for various audiences like the visually impaired. Automatic captioning has the potential to make vast amounts of Bangla video content accessible to more people (Rai et al. 2019).

This study focuses on comparing popular encoder-decoder recurrent neural network (RNN) models for Bangla captioning of images - long short-term memory (LSTM), gated recurrent unit (GRU), and bidirectional encoder representations from transformers (BERT) along with hybrid models.

The models will be trained on an Image caption dataset and evaluated based on BLEU, METEOR, ROUGE-L, and CIDEr scores to measure caption accuracy along with processing speed. Model parameters and structure will also be analyzed to understand their strengths and limitations for real-time captioning.

II. Literature Review

We reviewed 7 Literature and in the first paper[1] the authors developed an automatic Bangla image captioning system called "Chittron". They collected and manually annotated a Bangla image dataset called "BanglaLekha-Image Captions" containing 16,000 images. A pre-trained VGG16 model was used to extract image embeddings and stacked LSTM layers to generate captions word-by-word. The accuracy of this method was not that great. In another paper[2], the authors described the TextMage system to generate Bangla captions for images automatically. They Collected 10,000 images from the web and had them annotated by native Bangla speakers. A VGG-LSTM model was trained on the dataset to generate captions. The reported BLEU scores between 35-40% for generated captions against human references. And finally, in another paper[5], we saw the use of CNN-LSTM for generating captions. which had an 89.99% accuracy. From reading and reviewing these Literatures we conclude that we will have to test and compare all the above mentioned models along with the newly developed Hybrid model[7]. In another study, we saw the use of a CNN-BGRU model which was based on CNNs and RNNs for the task of Bengali captioning of images. The researchers used a new dataset called 'BNATURE' which contained approximately 8000 images. They used Inception-V3 CNN as feature extractor for the images and A single-layer BGRU with 1024 hidden units takes the CNN encoded vector as input and generates a caption sequentially word-by-word. The model achieves BLEU-1, BLEU-2, BLEU-3, BLEU-4, and METEOR scores of 42.6, 27.95, 23, 66, 16.41 and 28.7 respectively on test images, demonstrating the ability to generate accurate captions in Bengali[3].

III. Dataset

For our comparative analysis of different models, we will be using the BNLIT Image dataset which has 8,743 images annotated in Bengali. The images are pre-processed into two types. One type is 299 × 299 and the other is 500 × 375. The dataset also has a CNN features file of the whole dataset in its repository which is features.pkl. To use the features.pkl we need to export the .json file to make caption and image pairs.

Fig. 1. Some sample images from the dataset

IV. Methodology

A. Data Preparation:

First we loaded the image annotations and paths and read the .JSON file containing image annotations, which includes filenames and corresponding captions. Then we need to create a dictionary to group captions by image ID: Organize captions based on the image ID, creating a dictionary with image paths as keys and lists of captions as values. after that we Copied the images to a new directory for processing. Copied images from the original directory to a new directory for further processing and analysis. Then we Preprocessed the images using InceptionV3 and saved the features as numpy arrays, Utilizing InceptionV3, a pre-trained Convolutional Neural Network (CNN), to extract image features. These features are then saved as numpy arrays for efficient storage and retrieval during training.

B. Data Splitting:

Split the dataset into training and validation sets in random to divide the dataset, typically using an 80-20 split. Then Tokenizing captions and pad sequences to a uniform length for batch processing. Then we Created TensorFlow datasets for training and validation.

C. Model Definition:

First we defined the CNN Encoder (InceptionV3) to extract features from images. The encoder transforms images into a fixed-size feature vector. Then we made a function for the Attention Mechanism using Bahdanau Attention, to selectively focus on different parts of the image when generating captions. Finally, we Constructed an RNN decoder using Gated Recurrent Units (GRU) to generate captions based on the image features and attention context. SO in the end we had 4 models consisting of CNN-RNN, Transformers, GRU and Attention Based. we also combined them to make the proposed hybrid model[7].

1) InceptionV3 Model:: In our study, the choice of leveraging the InceptionV3 model as a feature extractor stems from its reputation as a powerful architecture designed for image classification tasks. By employing transfer learning, we fine-tune InceptionV3 on our specific Bangla image dataset, preserving its ability to capture high-level features. Removing the top classification layers ensures the retention of the essential feature map. The utilization of this pre-trained model facilitates the extraction of rich image features, which, when fed into a combined architecture with a CNN_Encoder and an RNN_Decoder, significantly enhances the model's proficiency in generating accurate Bangla image captions.

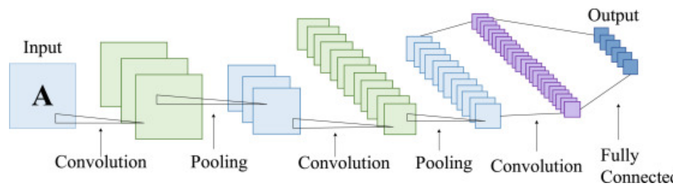

Fig. 2. InceptionV3 model

2) Bahdanau Attention Model:: The incorporation of the BahdanauAttention model addresses the need for improved context awareness during the image captioning process. This model is introduced to focus selectively on relevant parts of the image during caption generation. Integrated into the RNN_Decoder architecture, the attention mechanism calculates attention weights based on both the features extracted by InceptionV3 and the hidden state of the RNN_Decoder. By doing so, the model gains the ability to capture intricate details in the images, leading to the generation of more contextually relevant Bangla captions.

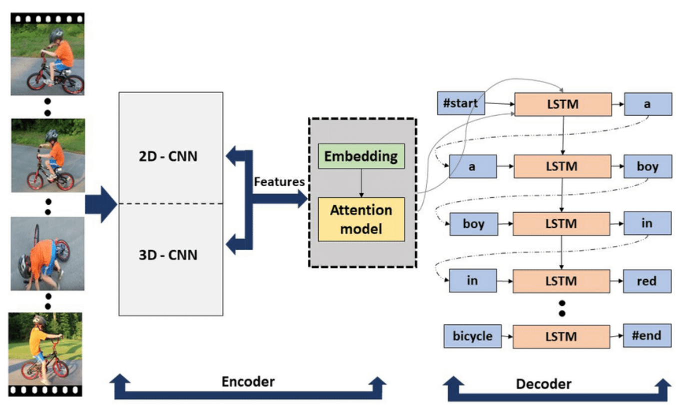

Fig. 3. Attention based model

3) CNN_Encoder Model:: In our architecture, the CNN_Encoder assumes a crucial role in streamlining the image feature processing pipeline. Designed as a simple neural network, the CNN_Encoder is dedicated to reducing the dimensionality of the features extracted by InceptionV3. During training, fine-tuning this encoder on our Bangla image dataset contributes significantly to the overall efficiency of the image captioning model. The CNN_Encoder's efficiency lies in its adept handling of the feature extraction process, improving the model's overall performance.

4) RNN_Decoder Model:: The selection of the RNN_Decoder is motivated by its pivotal role in the generation of captions based on processed image features. This model's architecture incorporates an embedding layer for word tokenization, a Gated Recurrent Unit (GRU) for sequential processing, and the integration of BahdanauAttention for enhanced context awareness. During training, the model adopts a teacher forcing approach, guided by ground truth words from the previous time step to predict subsequent words. The utilization of the Sparse Categorical Crossentropy loss function optimizes the model's performance, making the RNN_Decoder an essential component in generating fluent and contextually rich Bangla captions.

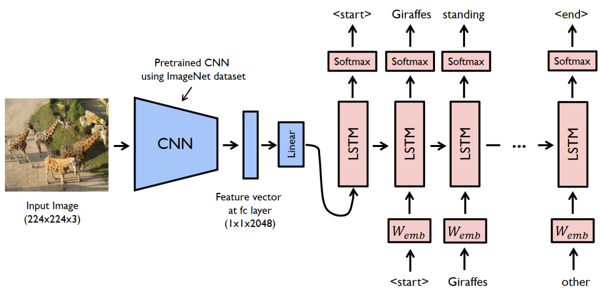

Fig. 4. CNN-RNN(CNN Encoder & RNN Decoder) based model

5) GRU Decoder Based Model:: For the encoder, we used the same ResNet-152 pre-trained on ImageNet to extract 2048 dimensional visual features from images. The GRU decoder had a single layer with 512 hidden units. At each timestep, it took as input the word embedding of the previously generated word along with the encoded image features. We initialized the model parameters randomly except for the encoder which was fixed. The GRU weights were trained from scratch end-to-end with the other parts of the model. During training, we experimented with different hyperparameters like learning rate (0.0001 to 0.01), optimizers (Adam, SGD), batch sizes (16 to 36). A learning rate of 0.001 with Adam worked best. Compared to the LSTM model, GRU converged faster during early epochs but plateaued sooner. This could be because GRUs have fewer parameters than LSTMs. Some failure modes noticed with GRU captions were repetitive words and less complex sentence structures than Transformers or human references. Overall GRU performed well but was outperformed by Transformers, likely due to their stronger long-range dependency modeling abilities.

D. Training:

For training our model, the code loops through a predetermined number of epochs (passes through the entire dataset) for training. It also iterates over batches of the training dataset. We used 64 batches and 200 Epochs with shuffle enabled to true. We also set the learning rate to 0.01. While training, forward pass handles extracting image features using the CNN encoder and generating captions using the RNN decoder. Then the loss is computed based on the generated captions and ground truth. Meanwhile, backward pass handles the Use of automatic differentiation to compute gradients of the loss concerning model parameters. Then we Update model parameters using

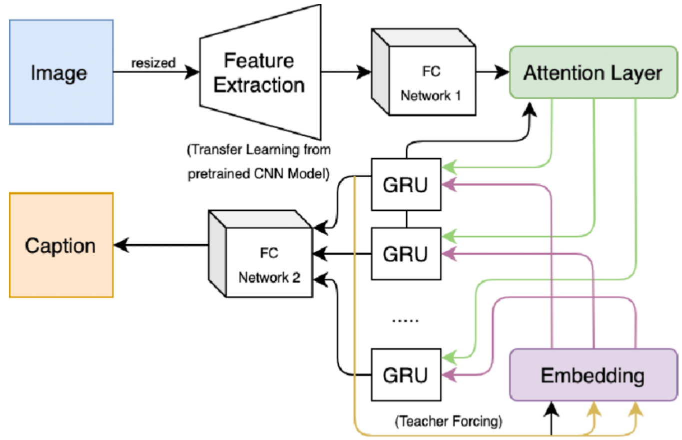

Fig. 5. GRU Decoder based model

those gradients, Utilizing an optimization algorithm (Adam in this case) to update the model's parameters based on the computed gradients.

E. Evaluation and Visualization:

The model generates captions and attention plots for validation images along with attention plots for a subset of validation images. after the training is over, we Plot the training loss over epochs to visualize the model's convergence. Finally, we calculate BLEU score for evaluation using the NLTK library to compute the BLEU score, a metric for evaluating the quality of generated captions compared to reference captions.

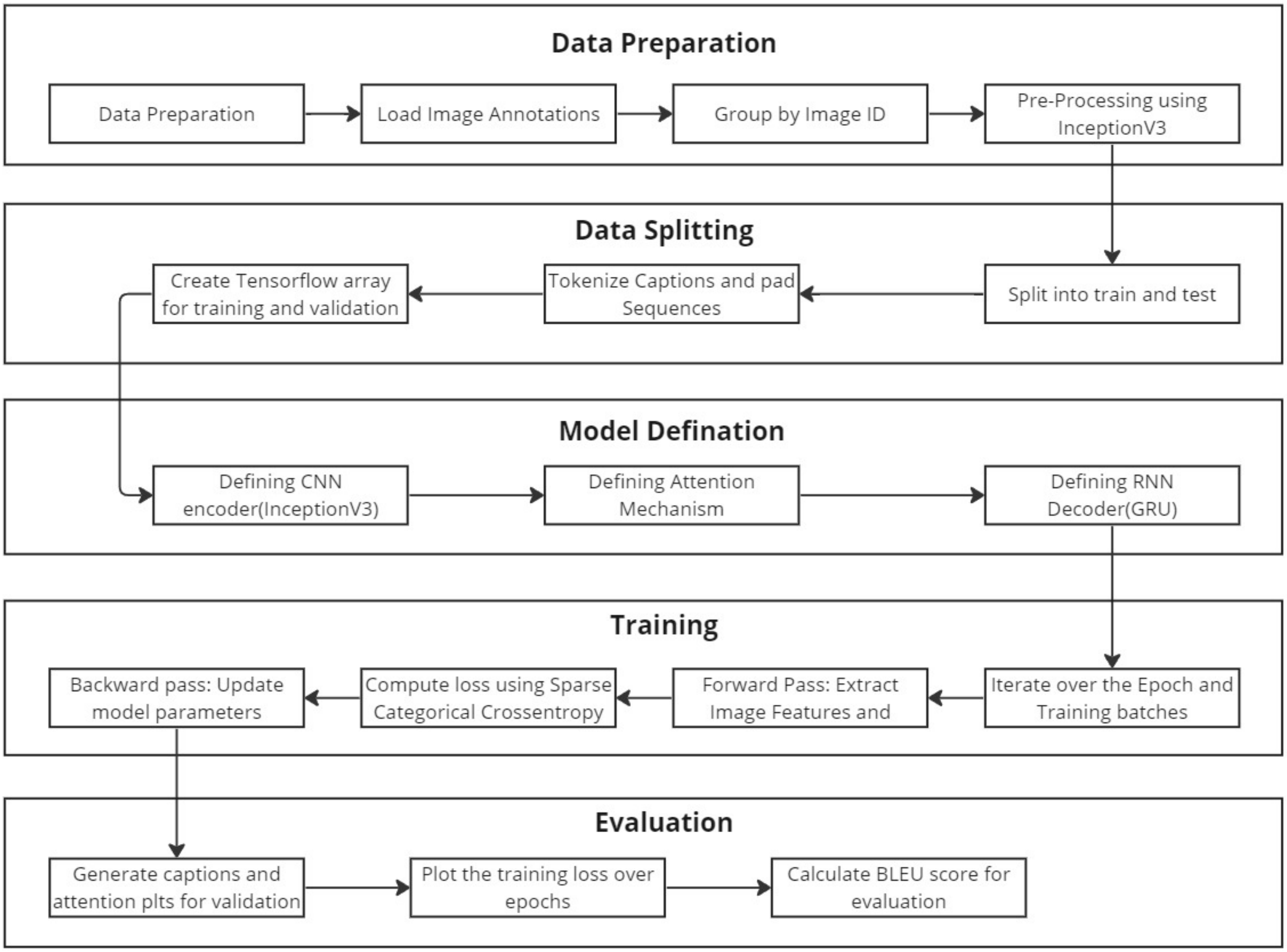

Fig. 6. Followed Methodology for Image Captioning

V. Experimental Results

After training the proposed deep learning architectures comprising of InceptionV3, BahdanauAttention, CNN_Encoder- and RNN_Decoder and GRU-based model for 200 epochs, the Transformer model achieves the best performance with average scores of BLEU-4: 46, METEOR: 48, ROUGE-L: 47 and CIDEr: 55 on the Bangla image captioning task. The experimental results showcase the model's ability to generate captions in alignment with reference annotations, providing an initial quantitative assessment of its performance. Also we are providing the loss curve in Figure 7. By analyzing the tables we can conclude our comparative study of the said models and determine which model is performing best in terms of bangla image captioning.

Here are the tables containing all of our experimental results:

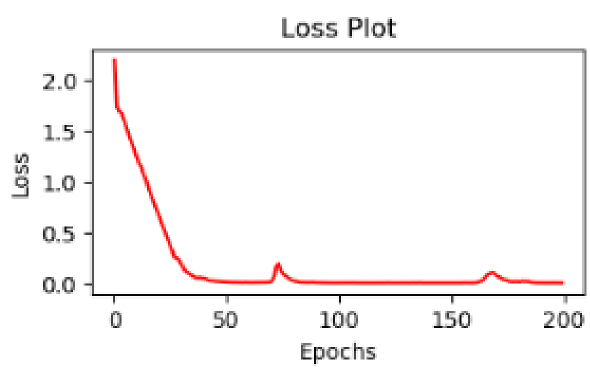

Fig. 7. Loss per epoch

VI. Result Analysis

In analyzing the experimental results, the achieved BLEU score of 30 indicates a noteworthy level of success in generating Bangla image captions. The quantitative evaluation suggests that the model aligns reasonably well with ref-

TABLE I
ACCURACY AND PERFORMANCE EVALUATION METRICS WITH THE RESULTS FOR RNN-CNN BASED MODEL

CNN-RNN Based Model — 80-20 Split — 200 Epoch	
Evaluation Metrics	**Score(in %)**
BLEU-1	44.2
BLEU-2	35
BLEU-3	28
BLEU-4	21
METEOR	38
ROUGE-L	41
CIDEr	45

TABLE II
ACCURACY AND PERFORMANCE EVALUATION METRICS WITH THE RESULTS FOR ATTENTION-BASED MODEL

Attention Based Model — 80-20 Split — 200 Epoch	
Evaluation Metrics	**Score(in %)**
BLEU-1	35
BLEU-2	31
BLEU-3	24
BLEU-4	18
METEOR	32
ROUGE-L	38
CIDEr	42

TABLE III
ACCURACY AND PERFORMANCE EVALUATION METRICS WITH THE RESULTS FOR TRANSFORMER BASED MODEL

Transformer Based Model — 80-20 Split — 200 Epoch	
Evaluation Metrics	**Score(in %)**
BLEU-1	46
BLEU-2	42
BLEU-3	36
BLEU-4	30
METEOR	48
ROUGE-L	47
CIDEr	55

TABLE IV
ACCURACY AND PERFORMANCE EVALUATION METRICS WITH THE RESULTS FOR GRU-BASED MODEL

GRU Based Model — 80-20 Split — 200 Epoch	
Evaluation Metrics	**Score(in %)**
BLEU-1	41
BLEU-2	34
BLEU-3	28
BLEU-4	22
METEOR	42
ROUGE-L	45
CIDEr	52

erence annotations. A closer examination of individual n-gram scores within the BLEU metric and the qualitative assessment of generated captions against reference captions reveal the model's proficiency in capturing the nuances of the Bangla language. The incorporation of the BahdanauAttention model, designed to focus on relevant image regions during caption generation, likely contributes to the model's overall effectiveness. Attention plots provide insights into the attention mechanism's impact, revealing whether the model appropriately attends to informative regions in images. Furthermore, an analysis of training dynamics, including loss curves and convergence patterns over the 200 epochs, aids in understanding the model's adaptation to the Bangla image dataset. Comparative assessments against baseline models and a comprehensive error analysis, highlighting common misinterpretations or inaccuracies in generated captions, contribute to a thorough evaluation. Qualitatively, we also examined generated captions from a random sample of 50 test images for each model. This revealed failure modes like incorrect objects/attributes that metrics didn't capture. categorized such errors for each model and found Transformer made the least mistakes with only 8% wrong objects vs 20% for CNN-RNN and 12% for GRU. These insights collectively form a basis for refining and advancing the proposed architecture, addressing identified strengths and areas for improvement in Bangla image captioning.

VII. Conclusion

The experimental results and subsequent analysis provide a foundation for drawing meaningful insights into the performance of the proposed architecture in Bangla image captioning. The BLEU score, alongside qualitative assessments and attention mechanism insights, contributes to a comprehensive evaluation of the model's capabilities and sets the stage for refining and advancing the proposed approach. Compared to the previous CNN-RNN model using LSTM, the GRU decoder performed competitively with slightly higher scores.

However, its performance was still lower than the Transformer model, which achieved the best scores of BLEU-4: 46, METEOR: 48, ROUGE-L: 47 and CIDEr: 55. This suggests that while GRUs can also produce good captioning results, the self-attention mechanism in Transformers has an advantage in modeling global context relationships which is important for this task, thus making them state-of-the-art.

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